

Antoinette Wangari

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• <https://github.com/AntoinetteWangari1>

• <https://www.linkedin.com/in/antoinettewangari/>

• <https://antoinettewangari1.github.io/PortfolioWebsite/>

Education

The Jomo Kenyatta University of Agriculture and Technology

Bachelor of Science in Geomatic engineering and Geospatial Information Systems

Certificate in Web and Mobile application design and development

Certificate in Data Science and Artificial Intelligence

Sept 2019 – Dec 2024

Jan 2019 – May 2019

Feb 2025 – May 2025

Relevant Coursework

- Spatial Data Science
 - Data Science
 - Remote Sensing
 - Chatbot development
 - Software Development
 - Artificial Intelligence
 - Internet of Things(IOT)
 - Computer vision
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Experience

GIS Consultant at StatsSpeak Group Limited

Feb 2024 – Present

- Worked on social impact mapping projects, including education and health dashboards, identifying gaps, mapping resources, and generating insights.
- Created Geonode dashboards for pharmaceutical mapping, aiding businesses in identifying competitors, market opportunities, and potential collaboration with health facilities.
- Created Python codes for route and delivery optimization for a delivery company, improving efficiency.
- Conducted geospatial business intelligence analysis for an online business, analyzing customer segments, product preferences, and market potential.
- Worked on Air Quality monitoring project to identify areas with better air quality aiding in risk reduction.
- Worked on Soil Moisture and Irrigation Monitoring of Bura Irrigation scheme to improve water management.

GIS Analyst at Kenya Agricultural and Livestock Research Organization

Jan 2023 – April 2023

- Worked as an attaché on a crop yield estimation and crop type classification system for small-scale farmers in Kenya using Machine Learning techniques.
- Collaborated with a team to develop a Web GIS application for the deployment of the crop yield estimation system, improving access to yield prediction for farmers.

Project Lead at Association of Geomatic and Geospatial Engineering Students

Oct 2023 – Oct 2024

- Worked with a team to organize GIS Day, securing sponsors, managing finances, planning the event, and ensuring its overall success.
- Collaborated with a team, including Digital Earth Africa and Measurement Systems, to organize upskilling events for students.
- Worked with a team, chairing weekly meetings to facilitate skill-sharing among GIS students and enthusiasts, fostering a community of learning.

Volunteer Mapper with Youth Mappers JKUAT chapter

Dec 2021 - Present

- Worked with Kenya Red Cross to map remote areas prone to floods, improving access for Red Cross volunteers.
- Collaborated with the Clinton Health Access Initiative to map schools, facilitating HPV vaccination campaigns aimed at preventing cervical cancer in female students.

Web Mapping Projects

- Developed a web map for Bura irrigation scheme to track water productivity and its irrigation efficiency.
https://antoinettewangari1.github.io/Bura_Irrigation_Scheme/
- Developed a web map to track the number of health facilities per ward to allow for resource mapping, and to show the different services offered by different health facilities.
<https://antoinettewangari1.github.io/Health-Facilities-in-Nairobi/>

Passion Projects

- Created a flood simulation of Tana River County using SuperMap.
https://www.linkedin.com/posts/antoinettewangari_i-recently-worked-on-a-hydrology-analysis-activity-7135535388705587200-Mpzz?utm_source=share&utm_medium=member_desktop&rcm=ACoAAD8red0BGxCEG_EumQL3DKTVVDJxUWCVow
- Conducted bathymetric survey at Juja Dam, to estimate the depth of the water body and the amount of siltation.
https://www.linkedin.com/posts/antoinettewangari_geospatialsience-bathymetry-watermanagement-activity-7229522908623118337-rg2e?utm_source=share&utm_medium=member_desktop&rcm=ACoAAD8red0BGxCEG_EumQL3DKTVVDJxUWCVow
- Built a smart flight recommender system to advice travelers on flights that offer the best value for money.
https://www.linkedin.com/posts/antoinettewangari_ai-datascience-machinelearning-activity-7325121297133252610-gSCo?utm_source=share&utm_medium=member_desktop&rcm=ACoAAD8red0BGxCEG_EumQL3DKTVVDJxUWCVow
- Created a farm monitoring IOT tool that monitors soil moisture and crop health and provides an advisory to farmers.
- Created an agricultural chatbot using RAG and OpenAI to advice farmers.
- Worked on a credit card fraud detection model.
- Created a model that conducts real time crop type and weed detection.

Technical Skills

- **Mapping Tools:** Leaflet, Openlayers, Geoserver, Geonode, QGIS, PostGIS & PostgreSQL, ArcGIS.
- **Languages:** JavaScript, Python, R.
- **Developer Tools:** Vs Code, Jupyter Notebook, Google Collab, TensorFlow, Keras

Extracurricular:

Mentorship: Geocode Connect Kenya Co-founder and Mentorship lead, training Geospatial students and GIS enthusiasts.